

1790 ▶ 1805

“I shall endeavor to find out how nature’s forces act upon one another”

Alexander von Humboldt, 1799

1792

All lit up

Scottish inventor William Murdoch invented gas lighting. He heated coal to produce a flammable gas and used it to light his home in Cornwall, England. Gaslights were much brighter than oil lamps, and the new form of lighting was soon being used to illuminate factories and streets.

Gas lamps reached London streets in 1809.



Nature travels
See pages 112–113

Scientific journey

German naturalist Alexander von Humboldt, who is regarded as the founder of ecology—the study of how organisms interact with the environment and each other—spent five years exploring South America from 1799 to 1804.



Humboldt’s woolly monkey is among the creatures he discovered.



1790

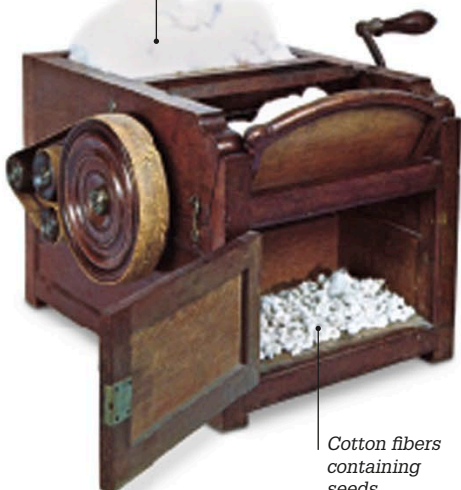
1795

1793

Cotton gin

In the US, inventor Eli Whitney invented the cotton gin, a mechanical device for removing the seeds from cotton fiber prior to spinning it, a job previously done by hand. The cotton gin led to a huge increase in the production of raw cotton.

Cleaned cotton



Cotton fibers containing seeds

1796 FIRST VACCINATION

English doctor Edward Jenner made a medical breakthrough with the first vaccination. He infected a healthy boy with cowpox, a disease similar to but milder than smallpox. When Jenner injected him with smallpox germs, the boy did not become ill—the dose of cowpox had built up his immunity. However, many people remained nervous about the process.



Blades used for vaccination

This cartoon of the time shows people sprouting cow’s heads after vaccination.

